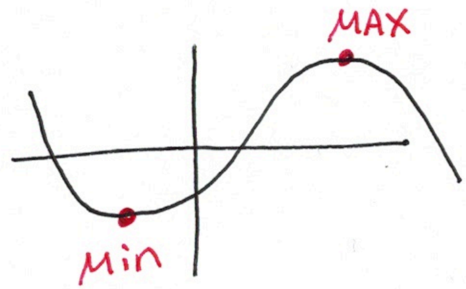
Section 4.7 Optimization



Ex ① Find two numbers whose diffeience is 100 and the product is a minimum.

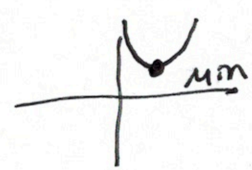
x and y

$x - y = 100 \rightarrow x = 100 + y$

$P = x \cdot y$

$P = (100 + y) \cdot y$

$P = 100y + y^2$



$P' = 100 + 2y$

$0 = 100 + 2y$

$-100 = 2y$

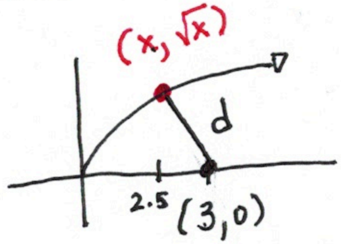
$-50 = y$

$x = 100 - 50$

$x = 50$

$\boxed{50 \text{ and } -50}$

Ex ② Find the point on $y = \sqrt{x}$ that is closest to $(3, 0)$.



$$d = \sqrt{(x-3)^2 + (\sqrt{x} - 0)^2}$$

$$d = \sqrt{x^2 - 6x + 9 + x}$$

$$d = \sqrt{x^2 - 5x + 9}$$

$$d' = \frac{2x - 5}{2\sqrt{x^2 - 5x + 9}} = 0$$

$$0 = 2x - 5$$

$$5 = 2x$$

$$2.5 = x$$

$$y = \sqrt{2.5} = 1.581$$

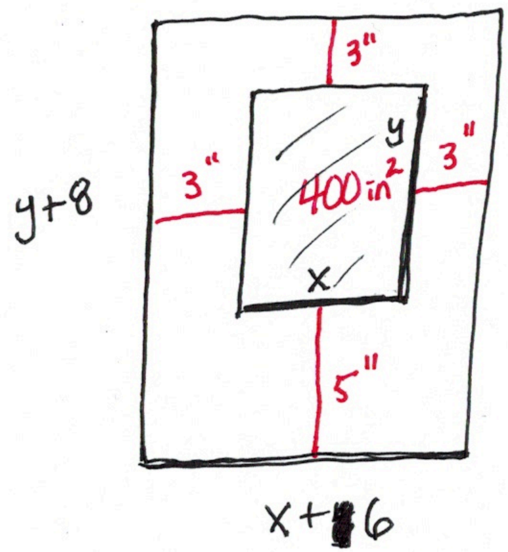
$$\boxed{(2.5, 1.581)}$$

Ex ③ Create a poster (rectangle) with the least paper possible.

The print area is $400\ in^2$.

The margins are 3" on top & left & right sides

The margin on the bottom is 5"



Auxillary Eq.

$$x \cdot y = 400$$

$$y = \frac{400}{x}$$

Main Eq.

$$P = (y+8)(x+6)$$

$$P(x) = \left(\frac{400}{x} + 8\right)(x+6)$$

$$P = 400 + \frac{2400}{x} + 8x + 48$$

$$P = 448 + \frac{2400}{x} + 8x$$

$$P' = -\frac{2400}{x^2} + 8$$

P' DNE $\cancel{x=0}$

$$0 = -\frac{2400}{x^2} + 8$$

$$\frac{2400}{x^2} = 8$$

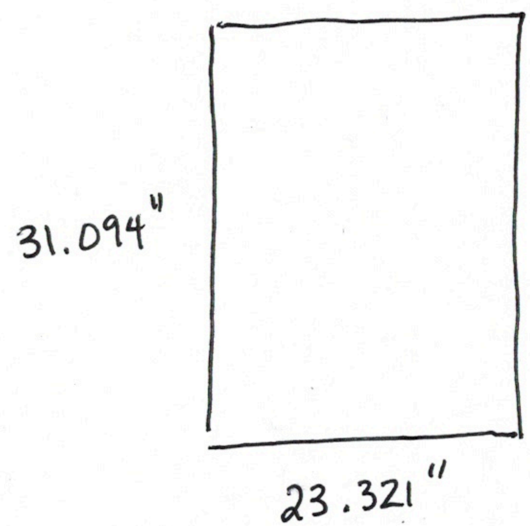
Ex 3 cont'd

$$\frac{2400}{8} = x^2$$

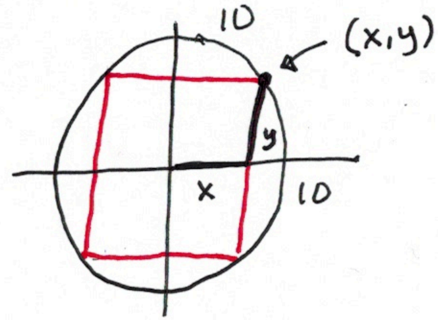
$$300 = x^2$$

$$x = 17.321$$

$$y = \frac{400}{17.321} = 23.094$$



Ex 4 Find the area of largest rectangle inscribed in a circle of radius 10".



$$x^2+y^2 = 100 \rightarrow y^2 = 100 - x^2$$

$$y = \sqrt{100-x^2}$$

Area of rectangle

$$A = (2x)(2y)$$

$$\boxed{50\,in^2} = \boxed{7.071'' \times 7.071''}$$

$$A = 4x\sqrt{100-x^2}$$

$$A' = 4\cdot\sqrt{100-x^2} + 4x\cdot\frac{-\cancel{2}x}{\cancel{2}\sqrt{100-x^2}}$$

$$\sqrt{100-x^2}\; 0 = 4\sqrt{100-x^2}\cdot\sqrt{100-x^2} - \frac{4x^2}{\sqrt{100-x^2}}\sqrt{100-x^2}$$

$$0 = 4(100-x^2) - 4x^2$$

$$0 = 400 - 4x^2 - 4x^2 \qquad y = \sqrt{100-7.071^2} = 7.071''$$

$$8x^2 = 400 \rightarrow x^2 = 50 \rightarrow x = \sqrt{50} = 7.071''$$